

# Inference-Compute Frontiers for Limit Order Book Prediction

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## Motivation

Short-horizon limit order book prediction is constrained by two different objects: predictive capacity and batch-one serving cost. A model that improves benchmark loss is not necessarily useful if the added computation is arranged in a way that is expensive under the target runtime. The paper studies this capacity-allocation problem directly on FI-2010: does a scaling-law-style frontier appear when loss is measured against inference compute capacity, and does that frontier survive contact with latency?

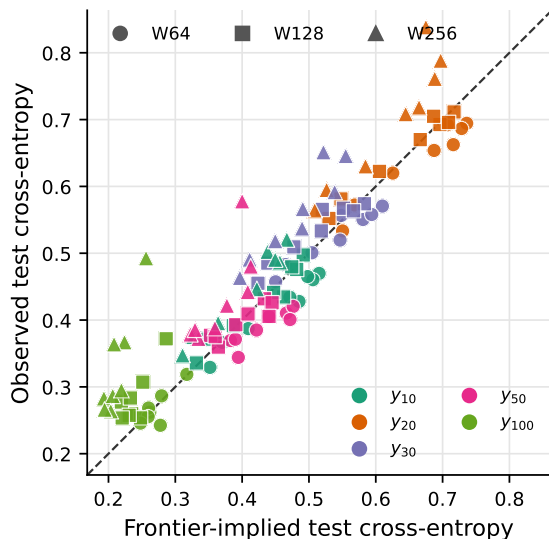
## Main empirical result

On raw LOB40 FI-2010 tasks, the realized lower envelope of held-out test cross-entropy versus structural forward work is well summarized by a power law:

$$L_*(C) = \min\{L(f) : \text{ops}(f) \leq C\}, \quad L(C) = L_\infty + AC^{-\alpha}.$$

The fit pool contains 2,520 datapoints across decision trees, histogram gradient boosting, CatBoost symmetric trees, and random-convolution logistic models. Additional model classes were screened, but did not attain the performance frontier. MLPLOB (the current published SOTA architecture for the FI-2010 benchmark) is held out as an architecture family and scored only after the non-MLPLOB frontier is fit. This model is 2 – 3 OOM larger than the fit models, testing whether or not the frontier curve can extrapolate into the large-model regime.

| diagnostic                | result        | interpretation                                                                                                               |
|---------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------|
| MLPLOB target frontier    | $R^2 = 0.941$ | Non-MLPLOB structural-work frontier extrapolates to held-out neural frontier points.                                         |
| W64+W128 MLPLOB subset    | $R^2 = 0.948$ | Properly fit high-compute targets are especially well explained.                                                             |
| All W64/W128/W256 targets | $R^2 = 0.838$ | W256 performs poorly even with more compute, so the claim is frontier regularity rather than monotone within-family scaling. |
| Latency-axis holdout      | $R^2 = 0.468$ | Wall-clock latency is not a drop-in substitute for structural work.                                                          |



**MLPLOB holdout.** Observed versus frontier-implied test cross-entropy for held-out W64/W128/W256 targets.

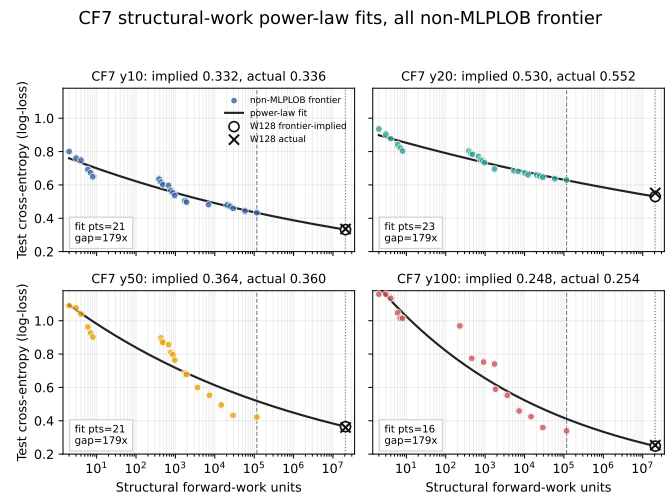
## Latency implication

Replacing structural work with measured CPU single-observation latency gives a much weaker holdout result and reorders model families: CatBoost has median structural work around 12,276 but latency around 281 $\mu$ s; histogram gradient boosting has lower median work, around 528, but latency around 4.34ms. W128 MLPLOB has about 20.85M structural-work units yet lower latency than several classical configurations.

## Constructive follow-up

This mismatch inspires the idea that for deployment viable models, one should cram the most compute capacity possible in latency-friendly computational structures. We develop the architecture FastBiNLOB in response. FastBiNLOB keeps normalization-aware, axis-aware LOB modeling, but replaces attention maps, Q/K/V projections, softmax attention, and repeated layout changes with dense temporal and feature MLP blocks.

FastBiNLOB is able to exceed currently published SOTA performance on some targets ( $y_{10}$  and  $y_{100}$ ) at notably lower latency (24% and 60% respectively). We do not claim SOTA performance across all horizons, but use this architecture as an example of how more useful compute can be arranged in a latency-friendly way.



**Example frontier.** Example lower-envelope points and fit power-law curve from CF7.